

Gate E5 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS (finite scope)**. Evidence directory: 20_EVIDENCE/E5/.

E5 predeclared tests and kill-switches (written 2026-09-30 before any E5 computation)

Evidence type for all: exact rational (Python `fractions`, integers) or exhaustive finite; exact ILP/LP only with exact certificate re-verification. Finite tests never establish universal ($\forall n$) claims. "FAIL" of a positive test $\Rightarrow$ suspend the claim and open a finding. "PASS" of a negative control is REQUIRED (the corrupted variant must be rejected); a negative control that is accepted invalidates the test.

ID	Claim (EN md line)	Test	Confirming outcome	Falsifying outcome	Limits
T01	(4.6a) l.380-386	exact eval at $n = 4e12$; monotone LHS; failure at $3.6e12$	holds at $4e12$, fails at $3.6e12$	holds at $3.6e12$ or fails at $4e12$	exact
T02	Lemma 4.2 l.366-368	$16 \cdot 117 / 1825 > 1$, $16 \cdot 12687 / 20000 > 10$	both true	either false	exact
T03	Lemma 5.3 coefficients l.672-688	recompute all four coefficient pairs from Table 2	equal to $-19/365$, $-253/500$, $-117/1825$, $-12687/20000$	mismatch	exact
T04	(5.12)-(5.14) l.618-655	exact evaluation of both coefficient bounds, (5.13) value, (5.12a)	$\leq 11/100$, $\leq 29/1000$, $\geq 393/100$, $< a/48$	any violated	exact; given the premises
T05	Thm 5.0 size window l.440	from $a \geq 33n/100$ (and $ n-3a \leq a/64$), $99a \leq 100p$, $48(2p-q) + \leq a$ derive $3267n \leq 10000p$, $3539p \leq 1188n$, $ 3p-n \leq n/50$	LP/rational derivation succeeds	a counter-assignment exists satisfying premises and violating a conclusion	exact LP over rationals (z3)
T06	(5.1)-(5.4) scalar steps l.456-514	$1e-12 + 1.5e-6 \leq (0.33)^2/65536$; $u(u+1)/2 \leq \varepsilon 0n^2 \Rightarrow u \leq 3n/(2 \cdot 10^6)$; $7a/2 - 129a/64 = 95a/64$; (5.1a) branch bounds vs $(2n+1)^2/24 - n^2/40$ for $n \geq 9$	all true	any false	exact / symbolic
T07	identity (6.3)/(6.8), (D.8), E.5 identity, (E.3) increments, (6.4) core sizes	exhaustive $n \leq 400$, all k	identities hold	any mismatch	finite exhaustive
T08	(6.7a) identity and $C(a+b, 2) \leq 10 \cdot d_R$ when $d_R > 0$; zero iff canonical; sharp at (3,2)	exhaustive $a, b \leq 60$	holds, (3,2) equality	violation; factor 9 (neg. control) must fail	finite
T09	(6.19) $\text{dist}(k, K)^2 \leq Q_s(n) - B_{\{s, n\}}(k)$	exhaustive $s \leq 20$, $n \leq 300$, admissible k	holds	violation	finite
T10	Prop 6.3a (6.20)-(6.21)	symbolic (sympy) + exhaustive small (s, q, d) ; brute-force rsd of $G_{\{s, q, d\}}$ for small; exact cp via ILP for tiny instances	formulas hold; $\text{rsd} \leq s$; $\text{cp} = c4 = Q_s - \delta_d$	any mismatch	ILP only tiny n
T11	E.3 witness $\text{rsd} \leq s$ and $\text{cp} = Q_s(n)$ (tiny n exact cp)	brute force	holds	fails	tiny n
T12	Lemma F.3a (F.5a)	exhaustive all graphs $n \leq 7$ (labelled up to iso via networkx atlas $n \leq 7$), all s with $\text{rsd} \leq s$, all D	holds	counterexample	finite; neg. control: replace s by $s-1$ must fail somewhere
T13	(F.7) $\text{indCopies}(C_k, G) \leq 2ks$ $n^{\{k-1\}}$	exhaustive $n \leq 7$	holds	violation	finite, weak test
T14	(F.9) chordal clique extraction	exhaustive chordal $n \leq 7$	holds	violation	finite
T15	Prop 6.4 (6.22), (6.23) $F_4 \leq \dots$	all graphs $n \leq 7$ (atlas) with $\text{rsd} \leq s$: compute F_4 exactly (LP + exact dual/primal certificate)	holds	violation	finite
T16	Prop D.1 $c4(K_n) \leq M(n)$ small n ; Prop D.2 K_n order ≤ 3 needs $> M(n)$ for $n \equiv 4 \pmod 6$	ILP small n (≤ 13) / parity argument check $n = 10$	D.1: packing found; D.2: ILP min with ≤ 3 pieces = $M(n) + 1$ at $n = 10$	otherwise	small n
T17	D.3 $W^*(S) = W(S) = 2C(k, 2)$ for K_{kvI_h} , $2 \leq k \leq h$	exact LP small k, h	equality	mismatch	small

T18	Thm C' constants: $\max\{40000(s+1)^3, 2/\varepsilon_s\} = 2e41(s+1)^8 \forall s \geq 0$; E.4 coefficient $1 + 32s + 800(3 + 32s)(s+1)^2 \leq 38000(s+1)^3$; total $\leq 40000(s+1)^3$	symbolic polynomial inequality over $s \geq 0$	true	false	exact
T19	F.1 deletion invariant algebra l.2141-2146	symbolic check with $\gamma_0 \leq \varepsilon/4, \varepsilon N_0 \geq 1$	invariant preserved	not	exact symbolic
T20	(E.2e) and (E.13) arithmetic; (E.12) finite core alternative	exhaustive small c,s (graphs on $c \leq 8$ vertices, $s \leq 2$) for CoreCliqueAlternative hypothesis $c \geq 20(s+1)^2$ is too large $\rightarrow$ test the combinatorial induction claim "missing graph max degree $\leq 2s$ and no matching $s+1 \Rightarrow \leq (4s+1)s$ edges" exhaustively on small graphs	holds	counterexample	finite
T21	Lemma 2.1/(1.3a) accounting; (2.4); Table 1	symbolic	ok	–	trivial
T22	Theorem 6.1/Cor 6.1a constants: $r + A + 3m \leq 488$ from (6.7); LP optimum of the account	exact LP	48 optimal from (6.7); 47 fails (neg. control)	otherwise	exact
T23	(G.4),(G.5)	exhaustive chordal K4-free $n \leq 8, e \leq 2n-3$; algebra of (G.5)	holds	violation	finite
T24	(6.5a)/(6.5) and Thm A arithmetic: $M(n) \geq n(n+1)/6 - 1$; $N^2 \leq 2^N$ for $N \geq 4$	exact	ok	–	trivial
T25	Certo	Only if a Certo certificate is bound in the target; else record NOT_APPLICABLE with reason	–	–	–

E5 record: independent falsification by exact computation

- **Gate:** E5, external adversarial audit of Paper IV v1.2, run `run_v1.2_r1`. Date: 2026-09-30.
- **Target:** `01_manuscript/v1.2_full_rebuild_candidate/PAPER_IV_preprint_v1.2_en.md`, sha256 `a501d53b527da22f20112db2b05917fb61f46dd8e080faf14b69349b240cc152`.
- **Predeclaration:** `E5_PREDECLARATION.md`, sha256 `e542a6566792bd4c996749db438c83847ed8956a231abaa4e340863a8b32e392`. It was written before any E5 computation and has not been modified.
- **Reading boundary:** only the manuscript, the predeclaration, the designated logging tool `tools/acclog.py` (with the header rows of `INPUT_ACCESS_LOG.csv`, read to confirm the format) and files created here. Every read is logged in `00_CONTROL/INPUT_ACCESS_LOG.csv`. No Lean sources, internal reports, author scripts or network resources were used.
- **Machine-readable output:** `E5_RESULTS.json`. Raw outputs are in `results/`, scripts in `scripts/` and run logs in `logs/`.

Verdict summary

24 PASS, 0 FAIL, 0 INCONCLUSIVE, 1 NOT_APPLICABLE (T25). Every predeclared negative control was rejected, as the predeclaration requires. T13 and T21 have no predeclared negative control. T15 has only an added power check, which is informational.

No computation falsified any claim tested. **No suspected mathematical defect was found.** The minor editorial observations below are not errors.

Method conventions

- A verdict is decided only in exact arithmetic: Python `int` and `Fraction`, sympy polynomial identities, or `z3` decision procedures for linear and nonlinear real arithmetic.
- **LP values W^* .** HiGHS proposes a solution. It is then rebuilt as exact rationals in one of two ways: `limit_denominator`, or, as a fallback, an exact Gaussian solve of the optimal simplex basis returned by HiGHS. Primal feasibility, dual feasibility and equal objective are all verified exactly. An instance without such a certificate would be INCONCLUSIVE. In the final runs, every LP (1253 in T15, 36 in T06s, 36 in T17) is exactly certified.
- **ILP values (cp, c_r) .** CBC returns a partition, which is re-verified exactly: every piece is a clique, of order ≥ 2 and $\leq r$, and every edge is covered exactly once. A value is labelled exact only if an independent exact

lower bound matches it. The bounds used are:

- an exhaustively checked weight (dual) certificate, where every clique has weight ≤ 1 (T10, T11);
- the parity argument (T16);
- $\lceil F_4 \rceil$ from an exact W^* (INFO_c4).

Otherwise the value is solver-dependent.

- **rsd.** Computed by brute force over all $U \subseteq V$ and all cliques $R \subseteq U$, taking the maximum over maximal cliques of $G[U]$, which is equivalent. Self-test: $\text{rsd} = 0 \Leftrightarrow$ chordal on all 1253 atlas graphs, with zero mismatches (`networkx.is_chordal`). Also $\text{rsd}(C4) = 1$ and $\text{rsd}(K_{\{3,3\}}) = 2$.

Per-test table

ID	Claim (EN md lines)	Method	Evidence	Range	Result	Negative control	Runtime (s)
T01	(4.6a) contraction at $n = 4e12$; fails at $3.6e12$ (380-386)	Fraction eval; sympy equivalence with $16D/n^2 < \epsilon 0 - 1/n$; derivative sign; exact crossover	exact	point + monotone	PASS	factor 20 rejected: PASS	0.4
T02	Lemma 4.2: $16 \cdot 117/1825 > 1$, $16 \cdot 12687/20000 > 10$ (366-368)	Fraction; LP max $(m + 10A)/D \approx 15.77 \leq 16$	exact	–	PASS	factor 15 rejected: PASS	0.0
T03	Lemma 5.3 coefficients from Table 2 (539-549, 672-688)	exact substitution chain; sympy (5.4a)	exact	–	PASS	corrupted 8A/40 rejected: PASS	0.0
T04	(5.12)-(5.14): $\text{coefA} = 0.10159 \leq 11/100$, $\text{coeff} = 0.02783 \leq 29/1000$, (5.13) = $3.930083 \geq 3.93$, (5.12a) (618-655)	Fraction; independent re-linearisation of (5.11)	exact, given the premises (5.12)	–	PASS	$R0 = 3.94$ and bound $1/10$ rejected: PASS	0.0
T05	Thm 5.0 size window (433, 440)	z3 LRA: premises $\wedge \neg \text{conclusion UNSAT}$	exact	real	PASS	$ 3p - n \leq n/100$ SAT; dropping $99a \leq 100p$ SAT: PASS	0.1
T06	(5.1)-(5.4) scalar steps, (5.1a) branches (456-514)	Fraction, sympy, z3 LRA/NRA, integer scan	exact/symbolic + finite scan	branch scan $n = 9..1500$	PASS (supp. T06s exact LP: PASS)	$1.7e-6$ and $n^2/8 - 1$ rejected: PASS	1.2 (+ 4.8)
T07	(6.3)/(6.8), (6.4), (D.8), E.5, (E.3), $Q_s \geq B$ (692-695, 759-775, 1785-1796, 1953-1979)	exhaustive integers	finite exhaustive	$n \leq 400$, all k in $[-n - 2, 2n + 2]$, $s \leq 60$	PASS	corrupted (6.3), (E.3) rejected: PASS	1.8
T08	(6.7a) identity; zero iff canonical; $C(a + b, 2) \leq 10d$; sharp (3,2) (862-877, 2253)	exhaustive integers	finite exhaustive	$a, b \leq 60$	PASS (unique equality at (3, 2))	factor 9 fails at (3,2): PASS	0.0
T09	(6.19) $\text{dist}(k, K)^2 \leq Q_s - B_{\{s, n\}}(k)$ (1004-1012)	exhaustive; $K = \text{argmax } B$ also checked	finite exhaustive	$s \leq 20$, $n \leq 300$, admissible and all k	PASS	$2 \cdot \text{dist}^2$ rejected (3302 cases): PASS	1.6
T10	Prop 6.3a (6.20)-(6.21) (1016-1052)	sympy; integer scan; brute-force rsd; ILP + weight certificate; brute-force min d_E over all labelled optimal templates	symbolic + finite + exact (tiny)	scan $s \leq 20$, $q \leq 200$; 11 instances $n \leq 16$	PASS	$\delta_d = d^2$ rejected; $\text{rsd} \leq s - 1$ rejected: PASS	52.5
T11	E.3 witness: $\text{rsd} \leq s$, $\text{cp} = Q_s(n)$ (1967-1979)	brute-force rsd; ILP cp and c3; exact weight certificate	exact per instance	$s \leq 3$, $2s + 2 \leq n \leq 14$	PASS ($\text{rsd} = s$, $\text{cp} = c3 = Q_s(n)$ in all 40 instances)	$\text{rsd} \leq s - 1$ rejected: PASS	20.4
T12	Lemma F.3a (F.5a) (2190-2209)	all atlas graphs, $s = \text{rsd}(G)$, all D	finite exhaustive	$n \leq 7$ (144,923 pairs)	PASS	$s - 1$ fails in 409 cases: PASS	0.3
T13	(F.7) induced C_k count (2261-2266)	all atlas graphs, all k	finite exhaustive (weak)	$n \leq 7$	PASS (max ratio 12/343)	none predeclared	0.1
T14	(F.9) chordal clique extraction (2275-2280)	all chordal atlas graphs, all D	finite exhaustive	$n \leq 7$, 532 chordal graphs	PASS	(added) factor 1 rejected: PASS	0.1
T15	Prop 6.4 (6.22), (6.23) (1058-1070)	exact W^* (certified LP) for all atlas graphs, all $s \in [\text{rsd}, n]$	finite exhaustive; LP exact	$n \leq 7$; 8974 / 1744 checks	PASS (min slack 0 for both)	(added power check) INFO	3.5
T16	Prop D.1 $c4(K_n) \leq M(n)$; Prop D.2 (1705-1761)	ILP constructions verified; parity lower bound	exact constructions; exact c3 values	D.1 $n \leq 13$; D.2 $n = 4, 10, 16$	PASS ($c3(K10) = 19$, $c3(K16) = 46$)	$n = 9$ (3 mod 6) rejected; $c3(K9) = 12 \leq 15$: PASS	140.7

T17	Prop D.3 $W^* = W = 2C(k,2)$ (1765-1781)	exact LP + explicit 1-factorisation packing	exact	$2 \leq k \leq 7, k \leq h, k + h \leq 13$	PASS	$k > h$ instances differ: PASS	1.0
T18	Thm C' constants; E.4 coefficient; $40000(s+1)^3$; $1+7A_s$ (101-107, 910-965, 2107-2118)	sympy (coefficients ≥ 0 after $x = 1 + y$); z3 chain for $s = 0..60$	exact	all $s \geq 0$ (symbolic)	PASS	$25000(s+1)^3$ rejected: PASS	0.1
T19	F.1 deletion invariant (2141-2146)	z3 NRA	exact	real	PASS	$\gamma_0 \leq \varepsilon$, and no $\varepsilon N_0 \geq 1$, both SAT: PASS	0.0
T20	(E.2e), (E.13) arithmetic; CoreCliqueAlternative counting step (1931-1940, 2076-2086)	exhaustive integers + z3; atlas graphs as missing graphs; SAT (CaDiCaL) $N = 8$	finite + exact; SAT solver-dependent	$c, b \leq 80, s \leq 15, t_W \leq 30$; graphs $n \leq 8$	PASS	15 in (E.13) SAT; $e \leq 2s$ rejected: PASS	4.5
T21	Table 1, (1.3), (1.3a), (2.1), (2.4) (113-138, 196-238)	symbolic + 3759 random verified partitions + all exact W^*	symbolic + finite	$n \leq 7$	PASS	none (trivial)	0.6
T22	Thm 6.1/Cor 6.1a: $r + A + 3m \leq 488$ optimal, $m + A \leq 16\delta$, 480 (837-877)	exact LP (vertex enumeration) + HiGHS	exact	–	PASS	47 rejected: PASS	3.5
T23	(G.4) $e \leq 2n - 3$; (G.5) $n \leq 20/\varepsilon$ (2390-2402)	chordal K4-free atlas + all simplicial extensions to $n = 8$; z3 NRA	finite exhaustive + exact	$2 \leq n \leq 8$	PASS (max $e = 2n - 3$ attained)	$2n - 4$ rejected; $19/\varepsilon$ SAT: PASS	0.1
T24	(6.5a)/(6.5), §2.3 floor, $N^2 \leq 2^*N$ (250, 301, 779-809)	exhaustive integers + sympy induction step	exact	$n \leq 1e5, N \leq 2000$ + induction	PASS	$N = 3$ and $-1/4$ rejected: PASS	0.3
T25	Certo (2414, 2444)	binding check	n/a	–	NOT_APPLICABLE	n/a	0

Supplementary checks (not predeclared; labelled as such)

- **S1.** Identity (6.7b) and inequality (6.16) (lines 872-877, 981-989) were tested on 4000 random graphs with $n = 4..9$. The clique partitions were random, of unrestricted order, and exactly verified. The roots were random cliques (4000 checks) and random template roots with all s (10,786 checks). **PASS.**
- **T06s.** Exact LP of $F_4(K_{r \vee I_h})$ against the (5.1a) branch bounds, for $4 \leq r \leq 11$ and $r + h \leq 12$. **PASS.**
- **INFO_c4.** c_4 was computed for all 1253 graphs with $n \leq 7$. 1132 values are exact via $\lceil F_4 \rceil$; 121 are solver-dependent optimal. Every upper bound is an exactly verified partition. Consequently $c_4(G) \leq Q_{\{rsd(G)\}}(n)$ holds exactly for every graph with $n \leq 7$. In particular, $c_4 \leq M(n)$ for every chordal graph with $n \leq 7$. This bears only on the open $b = 0$ question of §G.1 and does not bear on Theorem C, which is eventual.

Range reductions and deviations (declared)

- **T10.** Brute-force rsd , exact cp and the template-distance minimum were computed for 11 instances with $n \leq 16$ ($s \leq 2$). The closed-form claims were checked for $s \leq 20, q \leq 200, 1 \leq d \leq q/2$.
- **T11.** "Tiny n " was taken as $n \leq 14$ and $s \leq 3$.
- **T16 (D.2).** At $n = 16$ the second, infeasibility ILP solve was skipped because a first attempt exceeded the time budget (it hit the run's 595 s timeout, and CBC raised an error). $c_3(K_{16}) = 46 = M(16) + 1$ is still certified exactly, by the parity lower bound and a verified partition. At $n = 4$ and $n = 10$ both solves ran; the " $\leq M(n)$ " ILP was infeasible, which is solver-dependent.
- **T17.** Ranges $2 \leq k \leq 7, k \leq h, k + h \leq 13$.
- **T20.** The predeclaration asks for graphs with $c \leq 8$ vertices and $s \leq 2$.
 - $n \leq 7$ is covered exhaustively by the atlas.
 - $N = 8$ is covered by SAT (UNSAT for $s = 1, 2, 3$; $s = 3$ is trivially impossible). SAT UNSAT is solver-dependent.
 - For $s = 1$ the check is complete on all graph sizes: edges lie in a vertex cover of $\leq 2s$ vertices of degree $\leq 2s$, so there are at most 6 non-isolated vertices.
- **T06s (process note).** In the first T06s run, 2 of 36 LPs (K_{12} and $K_{10 \vee I_2}$) had no exact certificate from the `limit_denominator` step. The first script version mislabelled this as FAIL; under the protocol it should have been INCONCLUSIVE, and no bound was violated. An exact reconstruction from the HiGHS optimal basis was then added. This changes the verification method only; no threshold or claim was changed. The rerun certified all 36 LPs exactly and PASS. The rerun is the recorded result; the first run's console output was not kept.
- **Added negative controls.** Controls not in the predeclaration (T14 factor 1, T15 band power check, T20

$e \leq 2s$) are labelled "(added)" in the JSON.

Discussion of power and non-results

- **No FAIL and no INCONCLUSIVE** in the recorded results.
- **Low power in T13.** The bound (F.7) is extremely loose at $n \leq 7$ (maximum ratio 12/343). Beyond confirming that there are zero induced long cycles when $\text{rsd} = 0$, it has almost no falsifying power.
- **Low power in the T15 band condition.** At $n \leq 7$ the refined bound (6.23) was never violated even for s outside the band $4s \leq n$. The test therefore does not exercise the necessity of the band. It does exercise both bounds: the minimum slack is 0 for each.
- **Low power in T20.** The counting step is loose: the true maxima are 3 ($s = 1$) and 10 ($s = 2$) against 5 and 18. It also follows analytically, since $e \leq (2s)(2s) = 4s^2 \leq (4s + 1)s$.
- **(E.12) not tested at its hypothesis.** Its hypothesis $c \geq 20(s + 1)^2$ is out of exhaustive range. Without that hypothesis it fails for small c (24 cases with $c \leq 7$). This is expected and is not a finding.
- **T04 is conditional.** It verifies the arithmetic of (5.12)-(5.14) *given* the premises (5.12). The derivation of those premises from the regularisation estimates (5.4), (5.7) is a proof obligation; the manuscript lists it in A.2 as "not fully expanded". Finite computation cannot establish it.

Minor editorial observations (not defects)

1. **1.440.** The size window follows from $a \geq 33n/100$, $99a \leq 100p$ and $48(2p - q)_+ \leq a$ alone. $|n - 3a| \leq a/64$ is not needed; z3 confirms this. This is harmless.
2. **1.386.** The exact crossover of (4.6a) is $n^* = 3,672,542,735,043$. The statement "fails for $n \leq 3.6 \cdot 10^{12}$ " is correct but conservative.
3. **1.468.** $(4n + 1)^2/120$ is the unconstrained maximum of $(2b_0 - a_0)/3$. On the branch $a_0 \leq b_0 \leq 2a_0$ the maximum is about $(n^2 - 1)/8$. The stated bound is valid but loose.
4. **1.1953-1956.** (E.3), $Q_s(n) - Q_s(n - 1) = \lfloor (n + s + 1)/3 \rfloor$, holds for every $n \geq 1$ in the tested range, not only for sufficiently large n .
5. **1.2081.** Deleting both endpoints of a missing edge removes at most $4s - 1$ missing edges, not $4s + 1$. The stated count is an overcount and remains valid.

What these finite tests cannot establish

- They cannot establish any universal ($\forall n, \forall G$) claim: Theorems A, B, C, C', 6.1, 6.5, Proposition 6.4 for $n > 7$, or Lemma F.3a for $n > 7$.
- They say nothing about the existence or correctness of the asymptotic thresholds: N , N_s , N_s^{stab} , the tower bounds of §6.3 and Corollary 6.6, or (C.1)-(C.4).
- They do not cover the correctness of the rounding theorem (Thm 3.1, Appendix C), the localisation (Prop 4.3), the regularisation estimates (5.4)/(5.7), the normalisation estimates of Appendix E, or the removal argument of F.3/F.5.
- They do not cover the correspondence between the manuscript and the Lean sources.

Scalar and symbolic checks (T01-T06, T18, T19, T22, T24, and T05/T19 via z3) verify the stated arithmetic steps exactly for all real or integer values, *given their premises*. The small-graph checks can only falsify; within their ranges they found nothing to falsify.

E5 — lead-auditor verification

- Predeclaration (E5_PREDECLARATION.md) written by the lead auditor before any computation; the delegated implementation followed it (T01-T25).
- Reproduction: `t_scalar.py` and `t_integer.py` rerun by the lead auditor on 2026-09-30; outputs identical to the delegated results (excluding timings). Logs: `lead_rerun/`.
- Graph/LP/ILP tests (T10-T17, T15 LP certificates) were not rerun by the lead auditor (runtime); their outputs carry exact certificates re-verified in-script.
- Process deviation C-03/AUD-P1 recorded (lost first-run output of supplementary T06s).
- Status: **E5 PASS within finite scope** (24 PASS, T25 NOT_APPLICABLE: no Certo certificate is bound to any claim). Finite tests do not establish universal claims.